Project Requirements

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. Project Overview

Title: Trapped in Harry Potter: Rewrite the Fate
Platform: Android
Genre: Narrative Adventure

This project is an interactive narrative game where players attempt to change key events and save characters who died in the original story. The game follows a structured scene-based progression with deterministic scoring.

To enhance immersion and interactivity, the system integrates **LLM** and **RAG** technologies on top of a controlled game engine.

1. Functional Requirements

1.1 Narrative Gameplay

The system shall provide an interactive narrative experience based on predefined story scenarios.
Players shall be able to progress through a sequence of story scenes, each containing contextual descriptions and decision points.

1.2 Decision-Making System

The system shall allow players to make decisions at key points using predefined options (e.g., A/B/C).
The system shall evaluate each decision as correct, partially correct, or incorrect.
The system shall record the player's decision path throughout the game.

1.3 Scoring Mechanism

The system shall assign a score (survival points) based on player decisions.
Each decision shall contribute to the total score.
The final outcome shall be determined based on the accumulated score.

1.4 AI-Generated Narrative Feedback

The system shall generate dynamic narrative feedback after each player decision using a Large Language Model (LLM).
The feedback shall be context-aware and consistent with the story setting.
The system shall ensure that feedback enhances immersion without revealing future outcomes.

1.5 Free-text Persuasion Mechanism

The system shall allow players to input free-text responses in specific key scenarios.
The system shall analyze the input using an LLM and classify it into predefined strategy categories (e.g., tactical action, defensive action, invalid input).
The classification result shall influence the game outcome.

1.6 Knowledge Query System (RAG)

The system shall provide an in-game knowledge assistant (e.g., “Hermione’s Notes”).
Players shall be able to query background information about the game world.

The system shall retrieve relevant information from a knowledge base and generate responses using a Retrieval-Augmented Generation (RAG) approach.
The system shall provide hints without directly revealing correct decisions.

1.7 Ending System

The system shall determine the final outcome based on the player’s total score and key decisions.
The system shall generate a personalized ending narrative.
The system shall provide an AI-generated summary including:

- Key decisions made by the player
- Analysis of player decision style
- Explanation of the final outcome

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2. Non-functional Requirements

2.1 Performance

The system shall provide AI-generated responses within an acceptable time frame (e.g., 2–5 seconds).
The system shall ensure smooth scene transitions and interactions.

2.2 Usability

The user interface shall be intuitive and easy to navigate.
Players shall clearly understand available choices and game progression.

2.3 Reliability

The system shall ensure that core gameplay logic (e.g., scoring and branching) remains consistent and deterministic.
AI-generated content shall not affect core game rules.

2.4 Scalability

The system shall support future expansion through additional DLC scenarios.
The narrative structure shall be modular and extensible.

2.5 Security

API keys and sensitive data shall not be exposed on the client side.
The system shall implement secure communication between client and backend services.

2.6 Maintainability

The system shall separate rule-based logic from AI components.
Game content (e.g., scenes and choices) shall be stored in a structured and easily editable format (e.g., JSON).

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3. System Requirements

3.1 Platform

The system shall run on Android devices.

3.2 Backend Infrastructure

The system shall be deployed on a cloud platform (e.g., GCP), using services such as API Gateway, serverless or container-based services.

3.3 AI Integration

The system shall integrate with external LLM APIs for text generation and classification.
The system shall use embedding and retrieval mechanisms to support the RAG knowledge system.

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4. Constraints

- The project is limited by a fixed development timeline (Apr 5 – May 10).
- AI API usage may be constrained by cost and rate limits.
- LLM outputs are non-deterministic and must be controlled through prompt design and rule-based validation.
- The project is developed as a prototype/demo rather than a production-level system.

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5. Assumptions

- Users have basic familiarity with the Harry Potter universe.
- Users are willing to interact with text-based gameplay.
- A stable internet connection is available for AI-related features.